

Helix OTA — Issues (open workable items)

Revision: 2 **Last modified:** 2026-06-10T18:30:00Z

This is the canonical open-work tracker (§11.4.15 Status, §11.4.16 Type, §11.4.54 ATM-NNN). Closed items migrate to Fixed.md. The short-form companion is Issues_Summary.md. Items surfaced during the 2026-06-10 emulator-driven device-testing session; HEAD at creation a839220 + the docs commit on top.

§3. [ATM-003] Emulator Tier-2 — real Android A/B (update_engine/AVB/dm-verity auto-rollback) is host-gated

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - **WHAT:** Stand up the Tier-2 emulator — a real Android A/B update_engine payload-apply with AVB/dm-verity verification and auto-rollback, driven end-to-end against the control plane (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Tier-2 requires Cuttlefish (cvd) on a Linux host with nested KVM. Self-resolution exhausted (§11.4.21): (a) no host CLI/virtualisation path — the current host is Apple-Silicon with applehv, which cannot run Cuttlefish or nested KVM; (b) subagent delegation cannot conjure the missing kernel virtualisation; (c) the containers submodule (podman/applehv) cannot host a KVM-backed Cuttlefish guest on this hardware; (d) no captured fallback substitutes for real update_engine partition behaviour; (e) external research confirms Cuttlefish's Linux+KVM requirement — not a tooling gap. NOT structurally impossible (§11.4.112) — host/hardware-gated only. - **UNBLOCK CONDITION:** Access to a Linux host (or CI runner) with nested-KVM enabled where Cuttlefish (cvd) boots. - **WHO:** Operator — provision the Linux+KVM host / CI runner; see docs/design/EMULATED_DEVICE_TESTING.md Tier-2.

§4. [ATM-004] Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions

Status: Operator-blocked **Type:** Task

Operator-Block-Details: - WHAT: Tier-3 validation on the physical board — real vendor HAL, U-Boot A/B slot-switch, and dm-verity over real partitions, exactly as a fielded device runs (per docs/design/EMULATED_DEVICE_TESTING.md). - **WHY:** Requires the physical RK3588 / Orange Pi 5 Max hardware, which is not attached to this host. Self-resolution exhausted (§11.4.21): (a) no remote board over the available CLIs/ADB/SSH; (b) subagent delegation cannot substitute for absent silicon; (c) no repo tooling emulates real U-Boot slot-switch + on-silicon dm-verity; (d) no captured fallback reproduces real-partition AVB rollback; (e) research confirms the vendor HAL + bootloader path is board-specific. NOT structurally impossible (§11.4.112) — hardware-gated only. - **UNBLOCK CONDITION:** A physical RK3588 / Orange Pi 5 Max board reachable over ADB/SSH (or physically attached) for flashing. - **WHO:** Operator — attach / provide remote access to the board.

§5. [ATM-011] Workable-items SQLite database (constitution §11.4.93 mandate) not started

Status: In progress **Type:** Task **Description:** The constitution mandates a SQLite-backed single source of truth for workable items (docs/.workable_items.db, Go binary at cmd/workable-items/). Not started yet — no DB schema, no Go binary, no tracking.

§6. [ATM-012] Build resource stats tracking (constitution §11.4.24 mandate) not started

Status: In progress **Type:** Task **Description:** Constitution §11.4.24 mandates per-build resource telemetry (memory/CPU/IO). Not started. TSV registry + helper scripts need authoring.

§7. [ATM-013] Stress + chaos coverage gap (constitution §11.4.85 mandate)

Status: In progress **Type:** Task **Description:** Stress+chaos coverage exists for 2 of ~12 submodules (F86 — PARTIAL). Server endpoints now have stress+chaos (PR #13). Remaining submodules need coverage. Status: server DONE, submodule coverage growing.

§8. [ATM-014] Docs Chain submodule distribution (constitution §11.4.106 Phase 6) operator-gated

Status: In progress **Type:** Task **Description:** Docs Chain engine exists at / Volumes/T7/Projects/docs_chain with Phases 1-4 done and tested. Phase 6 (git submodule distribution) is operator-gated — needs explicit go-ahead to wire as a registered submodule. Meanwhile, .docs_chain/ YAML contexts + scripts/ sync_status_docs.sh are wired and working.